

User Manual: BaSyx Digital Product Pass (DPP) Web UI

Project 6: API for the Digital Product Pass (DPP) in the BaSyx Framework

Customer

Name	Mail
Markus Rentschler	rentschler@lehre.dhbw-stuttgart.de
Pawel Wojcik	pawel.wojcik@lehre.dhbw-stuttgart.de

Introduction

This document serves as the **User Manual** for the **BaSyx Digital Product Pass (DPP) Web UI**. It describes how end users can interact with and explore Digital Product Passes through the web-based interface. The interface is based on the DIN EN 18222 standard and is seamlessly integrated into the BaSyx Framework.

Live Web UI: <https://srv01.noah-becker.de/uni/swe/basyx/>

Document History

Version	Author	Date	Comment
1.0	Fabian Steiß	10.05.2026	Creation of the User Manual

Terms and Abbreviations

Abbreviation	Meaning
DPP	Digital Product Pass
AAS	Asset Administration Shell
BaSyx	Open Source Framework
Submodel	Part of an AAS
UI	User Interface

Table of Contents

1. [Purpose and Scope](#)
 1. [Purpose](#)
 2. [Scope](#)
2. [Use Cases](#)
3. [Web UI Features](#)
 1. [General Operation](#)

2. DPP-specific Features

4. Detailed Instructions
 5. Usability Concept & Workflows
 6. Tips and Troubleshooting
 7. MockUps & Screenshots
 8. References
-

1. Purpose and Scope

1.1 Purpose

This User Manual helps end users to easily and efficiently use the **Digital Product Pass** through the BaSyx Web UI. It explains all relevant functions of the DPP Viewer and shows how to navigate and view product information in a clear and structured way.

1.2 Scope

This manual refers exclusively to the **Web UI (Frontend)** available at <https://srv01.noah-becker.de/uni/swe/basyx/>.

It is intended for end users who want to view and explore Digital Product Passes.

2. Use Cases

The most important use cases from the user perspective are:

- Display and explore an existing Digital Product Pass
 - Navigate between different Submodels
 - View detailed product information
 - Switch between DPP View, AAS View, and Submodel View
-

3. Web UI Features

3.1 General Operation

FR-FE-05 – Main Menu

At the top of the application there is a menu that allows you to switch between the following views:

- **DPP Viewer**
- **AAS Viewer**
- **Submodel Viewer**

3.2 DPP-specific Features

FR-FE-01 – Visualize a Digital Product Pass

1. Open the Web UI

2. Switch to the **DPP Viewer**
3. Enter an **AAS ID**, **DPP ID**, or a full URL
4. The complete Digital Product Pass will be loaded

FR-FE-02 & FR-FE-03 – Navigation within the DPP

- A **sidebar** on the left side shows all available DPP Submodels
- Click on a Submodel to display its content in the main area

FR-FE-04 – Submodel Highlighting

The currently selected Submodel is visually highlighted in the sidebar.

FR-FE-06 – Product Information

The header area of the viewer displays:

- Product Name
- Product ID
- DPP Version
- Other key metadata

FR-FE-07 – Information Display

The data of each Submodel is presented in a clear **two-column layout**.

FR-FE-08 – Missing Data

If data is not available, it is clearly marked.

FR-FE-09 – Tooltips

Hover over individual data fields to see explanatory tooltips based on DIN EN 18222 and IDTA specifications.

FR-FE-10 – Responsive Design

The interface automatically adapts to desktop and mobile devices.

4. Detailed Instructions

How to Display a DPP

1. Open the Web UI
2. Select **DPP Viewer** in the top menu
3. Enter the desired ID or URL and confirm
4. Use the left sidebar to browse through the Submodels

Switching Submodels

Simply click on the desired Submodel in the left sidebar (e.g. Digital Nameplate, Product Carbon Footprint, Material Composition, etc.).

5. Usability Concept & Workflows

Core Workflow:

1. Find a product / DPP
2. Switch to the **DPP Viewer**
3. Select the desired Submodel via the sidebar
4. View the information and use tooltips for more details

Additional features:

- Direct deep links to specific DPPs
 - Clear visual highlighting and separation
 - High readability thanks to the two-column layout
-

6. Tips and Troubleshooting

Problem	Solution
DPP does not load	Verify the ID and ensure the AAS exists in the system
Submodels are missing	The DPP must contain the corresponding IDTA Submodels
Display is incomplete	Reload the page or try another browser (Chrome/Firefox recommended)
No tooltips appear	Hover the mouse over the field labels

7. MockUps & Screenshots

Screenshots of the current interface can be found directly in the Web UI or in the project repository under [/PROJECT/PM/Mockups & Wireframes](#).

8. References

No.	Reference	Title
1	DIN EN 18222	Digital Product Passport - APIs
2	IDTA-02035-x	Submodel Templates for DPP
3	BaSyx Framework	Eclipse BaSyx
4	HARTING Demo	Reference Implementation